Demo_4_PopulationStratification_2025

Cris Van Hout

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```r
knitr::opts_chunk$set(echo = FALSE)
knitr::opts_chunk$set(message = FALSE)
knitr::opts_chunk$set(warning = FALSE)

# Load relevant libraries
require(ggplot2)
```

```
## Loading required package: ggplot2
```

```r
#Define parameter values for two possibly distinct populations
##p, major allele frequency
##mean, sample mean value from a normally distributed variable
##sd, standard deviation value from a normally distributed variable
##effect, the beta coefficient of the minor allele
##numObs, the sample size
##all parameters can be set separately for populations 1 and 2

p1=.5
mean1=0
sd1=1
effect1=0
numObs1=500
numObs2=500
p2=.9
mean2=1
effect2=0
sd2=1

# Note that as sample sizes go up, so does sensitivity to differences in
#effect size and allele frequency, here we consider a low sample size
#with a large genotype frequency difference as an illustration

#Note: coding the genotypes this way is different than the last demo
hwprops1=as.integer(as.character(numObs1*(c(p1^2,2*p1*(1-p1),(1-p1)^2))))
hwprops2=as.integer(as.character(numObs2*(c(p2^2,2*p2*(1-p2),(1-p2)^2))))

geno1<-c(rep(0,hwprops1[1]),rep(1,hwprops1[2]),rep(2,numObs1-hwprops1[1]-hwprops1[2]))
pheno1<-rnorm(numObs1,sd=1,mean=(mean1+effect1*geno1))
geno2<-c(rep(0,hwprops2[1]),rep(1,hwprops2[2]),rep(2,numObs2-hwprops2[1]-hwprops2[2]))
pheno2<-rnorm(numObs2,sd=sd2,mean=(mean2+effect2*geno2))
genoall<-c(geno1,geno2)
```

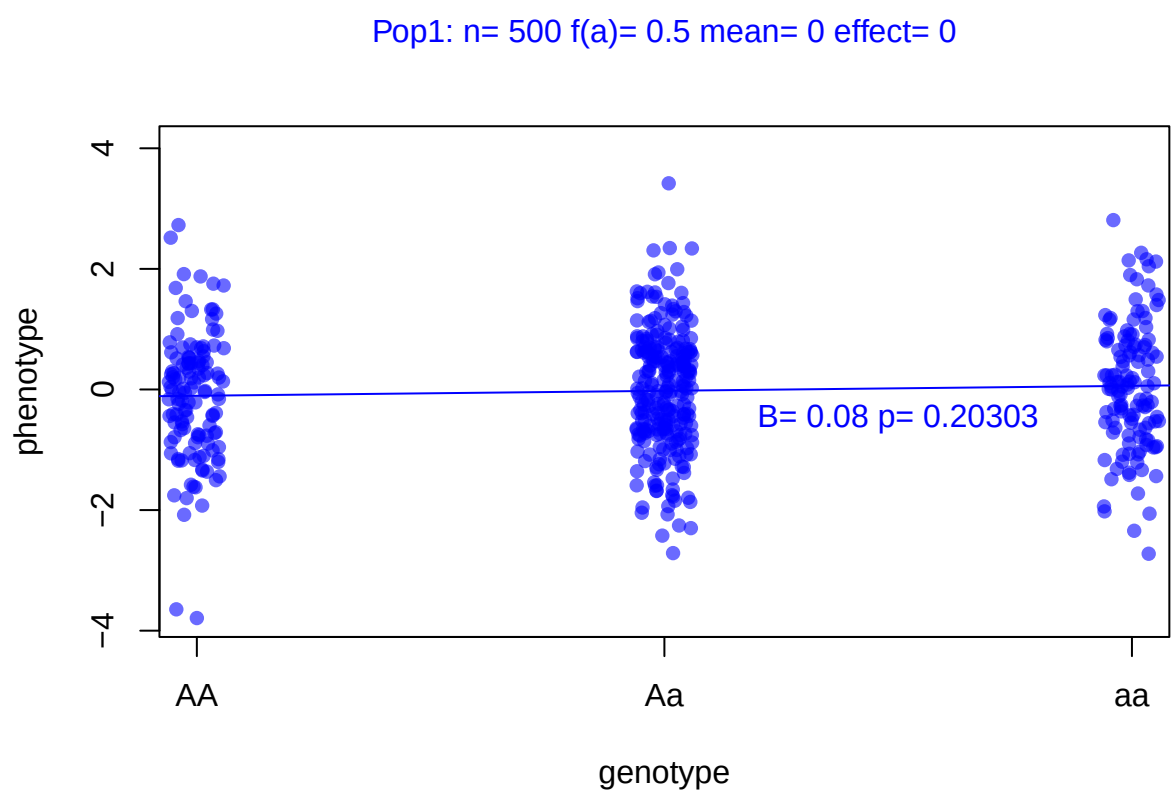
```
phenoall<-c(pheno1,pheno2)
xlim=range(genoall)
ylim=range(phenoall)
ancestry<-c(rep("Blue",numObs1),rep("Red",numObs2))


#consider the effect of the genotype in population 1
titlept1=paste("Pop1: n=",numObs1,"f(a)=",(1-p1),"mean=",mean1,"effect=",effect1)
plot(jitter(geno1,.3),pheno1,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(0,0,255,150,ma
mtext(titlept1, side = 3, line = 2, cex = 1, col = "blue")
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)

lm1<-lm(pheno1~geno1)
sum.lm1<-summary(lm1)
pval1<-print(sum.lm1$coefficients[2,4],digits=2)
```

```
## [1] 0.2
```

```
beta1<-sum.lm1$coefficients[2,1]
abline(lm1$coefficients[1],lm1$coefficients[2],col="blue")
label1<-paste("B=",round(beta1,2),"p=",round(pval1,5))
text(1.5, y = ((lm1$coefficients[1]+1.5*lm1$coefficients[2])-.5), labels = label1,col="blue")
```

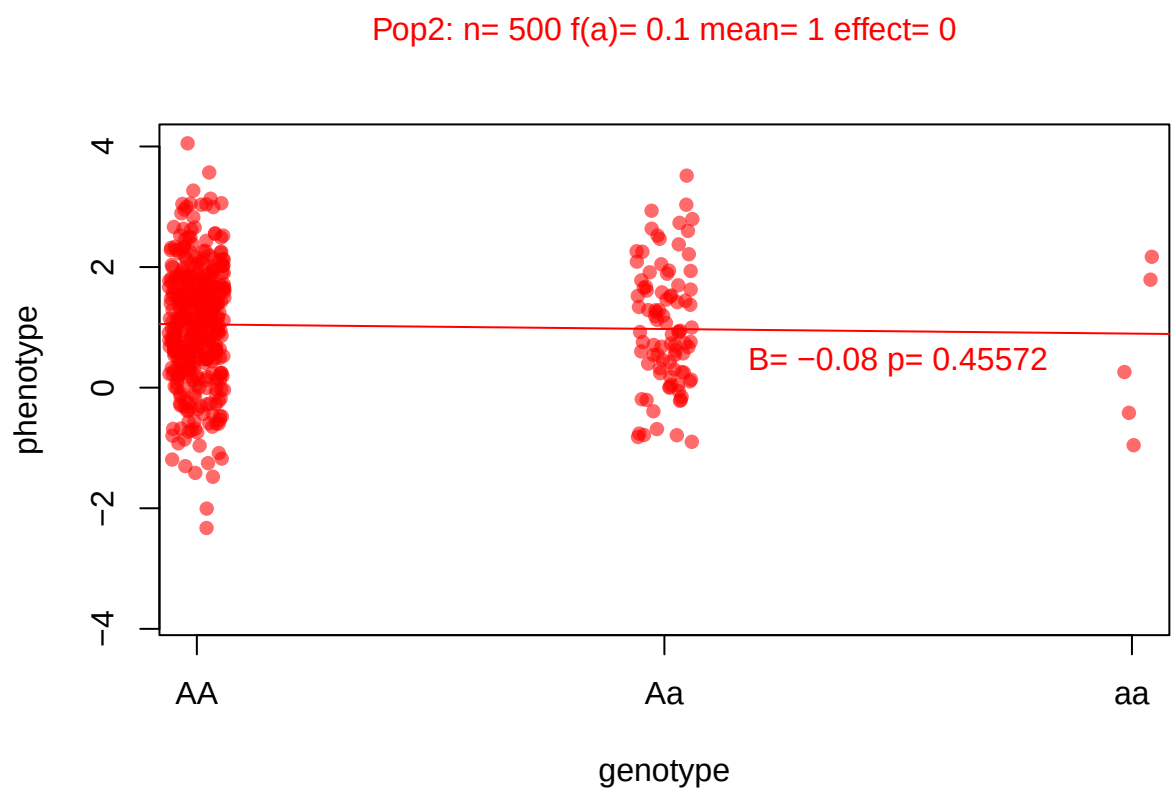


```
#consider the effect of the genotype in population 2
titlept2=paste("Pop2: n=",numObs2,"f(a)=",(1-p2),"mean=",mean2,"effect=",effect2)
plot(jitter(geno2,.3),pheno2,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(255,0,0,150,ma
mtext(titlept2, side = 3, line = 2, cex = 1, col = "red")
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)

lm2<-lm(pheno2~geno2)
sum.lm2<-summary(lm2)
pval2<-print(sum.lm2$coefficients[2,4],digits=2)
```

```
## [1] 0.46
```

```
beta2<-sum.lm2$coefficients[2,1]
abline(lm2$coefficients[1],lm2$coefficients[2],col="red")
label2<-paste("B=",round(beta2,2),"p=",round(pval2,5))
text(1.5, y = ((lm2$coefficients[1]+1.5*lm2$coefficients[2])-.5), labels = label2,col="red")
```

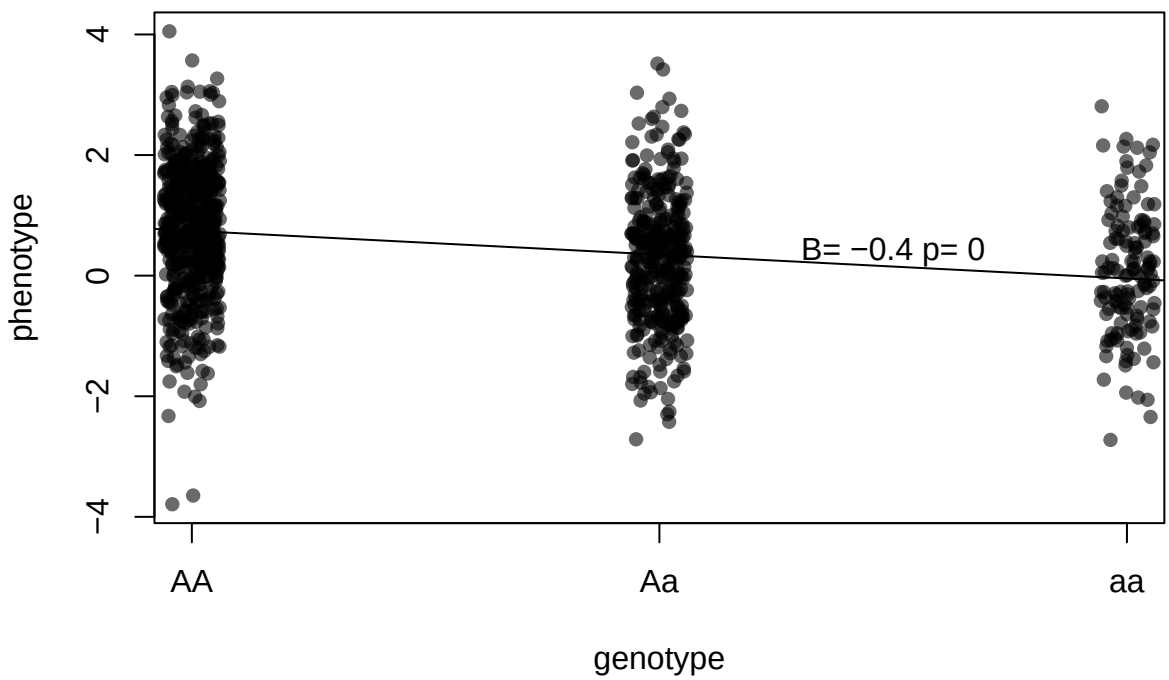



```
#consider the effect of the genotype in population 1 and 2 jointly, ignoring any difference
## A significant effect of the a allele is observed,
##despite having simulated no effect of the genotype in either population 1 or 2
plot(jitter(geno1,.3),pheno1,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(0,0,0,150,max
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)
```

```
points(jitter(geno2,.3),pheno2,col=rgb(0,0,0,150,maxColorValue=255), pch=16)

lmall<-lm(phenoall~genoall)
sum.lmall<-summary(lmall)
pvalall<-sum.lmall$coefficients[2,4]
betaall<-sum.lmall$coefficients[2,1]

abline(lmall$coefficients[1],lmall$coefficients[2])
labelall<-paste("B=",round(betaall,2),"p=",round(pvalall,6))
text(1.5, y = ((lmall$coefficients[1]+1.5*lmall$coefficients[2])+.25), labels = labelall)
```




```
#Whoa! The model that ignores that there are actually two different groups
#identifies a significant effect of the a allele when we coded no effect
#in both populations! This is an illustration of a false positive result
#caused by population stratification, differences in allele frequency and
#mean trait value between two populations in the absense of any true effect
#of the allele!


#consider the effect of the genotype in population 1 and 2 jointly, accounting for difference
titlept1=paste("Pop1: n=",numObs1,"f(a)=",(1-p1),"mean=",mean1,"effect=",effect1)
titlept2=paste("Pop2: n=",numObs2,"f(a)=",(1-p2),"mean=",mean2,"effect=",effect2)
title12=c(titlept1,titlept2)
plot(jitter(geno1,.3),pheno1,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(150,0,0,150,ma
mtext(titlept1, side = 3, line = 2, cex = 1, col = "red")
```

```r
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)

points(jitter(geno2,.3),pheno2,col=rgb(0,0,150,150,maxColorValue=255), pch=16)
mtext(titlept2, side = 3, line = 1, cex = 1, col = "blue")

lmall<-lm(phenoall~genoall)
sum.lmall<-summary(lmall)
pvalall<-print(sum.lmall$coefficients[2,4],digits=6)
```

```
## [1] 1.75105e-15
```

```r
betaall<-sum.lmall$coefficients[2,1]

lm1<-lm(pheno1~geno1)
sum.lm1<-summary(lm1)
pval1<-print(sum.lm1$coefficients[2,4],digits=3)
```

```
## [1] 0.203
```

```r
beta1<-sum.lm1$coefficients[2,1]

lm2<-lm(pheno2~geno2)
sum.lm2<-summary(lm2)
pval2<-print(sum.lm2$coefficients[2,4],digits=3)
```

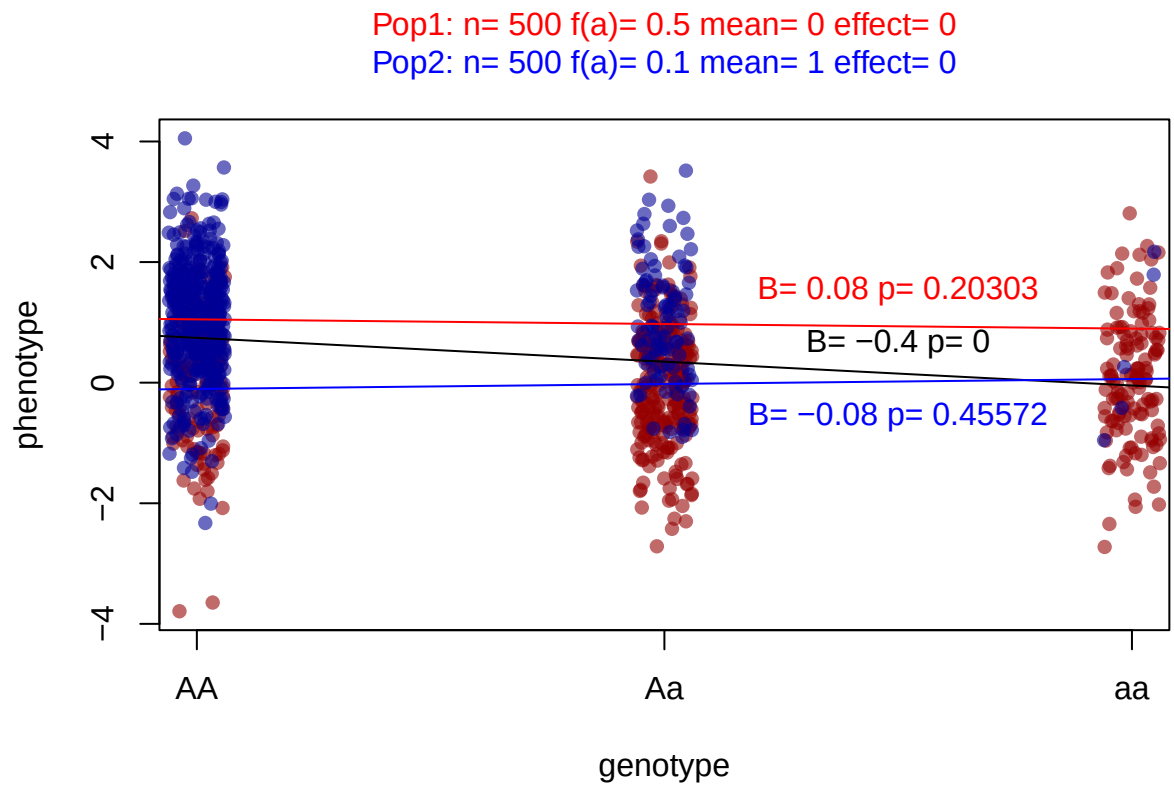
```
## [1] 0.456
```

```r
beta2<-sum.lm2$coefficients[2,1]

abline(lmall$coefficients[1],lmall$coefficients[2])
labelall<-paste("B=",round(betaall,2),"p=",round(pvalall,5))
text(1.5, y = ((lmall$coefficients[1]+1.5*lmall$coefficients[2])+.5), labels = labelall)

abline(lm1$coefficients[1],lm1$coefficients[2],col="blue")
label1<-paste("B=",round(beta1,2),"p=",round(pval1,5))
text(1.5, y = ((lm1$coefficients[1]+1.5*lm1$coefficients[2])+1.5), labels = label1,col="red")

abline(lm2$coefficients[1],lm2$coefficients[2],col="red")
label2<-paste("B=",round(beta2,2),"p=",round(pval2,5))
text(1.5, y = ((lm2$coefficients[1]+1.5*lm2$coefficients[2])-1.5), labels = label2,col="blue")
```




```
#Ok, now that we see the issue. Lets build a model that resolves the problem
##Recall, population 1, true effect of a allele is zero
sum.lm1
```

```
##
## Call:
## lm(formula = pheno1 ~ geno1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.6864 -0.6638  0.0406  0.6498  3.4408
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.10404    0.07896  -1.318    0.188
## geno1        0.08218    0.06447   1.275    0.203
##
## Residual standard error: 1.019 on 498 degrees of freedom
## Multiple R-squared:  0.003252,   Adjusted R-squared:  0.00125
## F-statistic: 1.625 on 1 and 498 DF,  p-value: 0.203
```

```
##Recall, population 2, true effect of a allele is zero
sum.lm2
```

```
##
```

```
## Call:
## lm(formula = pheno2 ~ geno2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.3763 -0.5894  0.0306  0.6402  3.0023
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.04953    0.04805  21.842   <2e-16 ***
## geno2       -0.07647    0.10244  -0.747    0.456
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9719 on 498 degrees of freedom
## Multiple R-squared:  0.001118,   Adjusted R-squared:  -0.000888
## F-statistic: 0.5573 on 1 and 498 DF,  p-value: 0.4557
```

```r
##Recall, both populations, not accounting for ancestry, we observe a spurious false positive
#apparent significant effect of a allele
sum.lmall
```

```
##
## Call:
## lm(formula = phenoall ~ genoall)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.5337 -0.7337  0.0412  0.7499  3.3086
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.74325    0.04530  16.406  < 2e-16 ***
## genoall     -0.39511    0.04885  -8.088 1.75e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.092 on 998 degrees of freedom
## Multiple R-squared:  0.06151,    Adjusted R-squared:  0.06057
## F-statistic: 65.41 on 1 and 998 DF,  p-value: 1.751e-15
```

```r
##Finally, both populations including the confounding variable 'ancestry'
##in the linear model to account for population stratification
##The apparent effect of a allele is estimated to be near the true effect, zero
lmallA<-lm(phenoall~genoall+ancestry)
summary(lmallA)
```

```
##
## Call:
## lm(formula = phenoall ~ genoall + ancestry)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -3.7284 -0.6479  0.0369  0.6470  3.4408
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.06205    0.07003  -0.886    0.376
## genoall      0.04019    0.05403   0.744    0.457
## ancestryRed  1.08824    0.07641  14.242   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9963 on 997 degrees of freedom
## Multiple R-squared:  0.2202, Adjusted R-squared:  0.2186
## F-statistic: 140.7 on 2 and 997 DF,  p-value: < 2.2e-16
```

```
#Summary: As we see in the parameters of the simulation, it's is not the genotype
#that is associated with the outcome, it is the ummodeled difference in ancestry.

#Question: How could we account for differences in ancestry without pre-defined
#ancestry groups?
```